2. Differentiate between Supervised and Unsupervised Learning

Feature	Supervised Learning	Unsupervised Learning
1. Definition	A type of machine learning where the model is trained on a labeled dataset (input-output pairs) to predict future outcomes.	A type of machine learning where the model identifies hidden patterns or structures within unlabeled data.
2. Goal	To map an input (x) to an output (y) based on provided examples.	To find the underlying structure or distribution in the data.
3. Supervision	Operates under the guidance of a "teacher" or labeled data.	Operates on its own to discover information.
4. Types	Categorized into Regression and Classification.	Categorized into Clustering and Association.
5. Accuracy	Highly accurate as it uses verified labels for training.	Less accurate; results are subjective and depend on pattern

		discovery.
6. Examples	**Spam detection (Gmail), House price prediction.**	**Customer segmentation (Netflix), Image compression.**

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3. Informed vs. Uninformed Search Algorithms

Feature	**Uninformed Search (Blind Search)**	**Informed Search (Heuristic Search)**
1. Definition	A search strategy that explores the search space **without any domain knowledge** or information about the distance to the goal.	A search strategy that uses **heuristic functions** (additional information) to estimate the cost/distance to the goal.
2. Knowledge	Knows only the start state, the goal state, and valid operators.	Uses domain-specific knowledge to guide the search towards the goal efficiently.
3. Performance	Generally slower and consumes more memory (explores all directions).	Faster and more memory-efficient as it focuses on the most promising paths.
4. Heuristic ($h(n)$)	Does **not** use a heuristic function.	Uses a heuristic function $h(n)$ to rank nodes.

5. Complexity	High time and space complexity.	Lower complexity due to targeted exploration.
6. Examples	Breadth-First Search (BFS), Depth-First Search (DFS).	A* Search, Greedy Best-First Search.

Comparison of Forward and Backward Chaining

Feature	Forward Chaining	Backward Chaining
Approach	Data-driven	Goal-driven
Starting Point	Known facts	Specific goals
Efficiency	Efficient for exploring all possible inferences	Efficient for achieving specific goals
Memory Usage	Can be memory intensive due to storing intermediate facts	Typically requires less memory
Implementation Complexity	Simple to implement	More complex to implement due to recursive nature
Suitability	Suitable for dynamic environments with continuously arriving data	Suitable for diagnostic systems and interactive applications
Handling Large Rule Sets	May become slow with large rule sets	Can be complex to manage with large rule sets

Feature	Propositional Logic	First-Order Logic
Basic Unit	Propositions	Predicates, constants, variables
Expressiveness	Limited to true/false statements	Expressive, can represent relationships and properties
Quantifiers	None	Universal (∀) and Existential (∃)
Syntax	Combines propositions using logical connectives	Uses predicates and quantifiers
Semantics	Truth tables	Interpretation over a domain
Use Cases	Simple problems (e.g., circuit design, rule-based systems)	Complex problems (e.g., AI reasoning, ontology modeling)
Example	P→Q	$\forall x \exists y (Likes(x, y))$